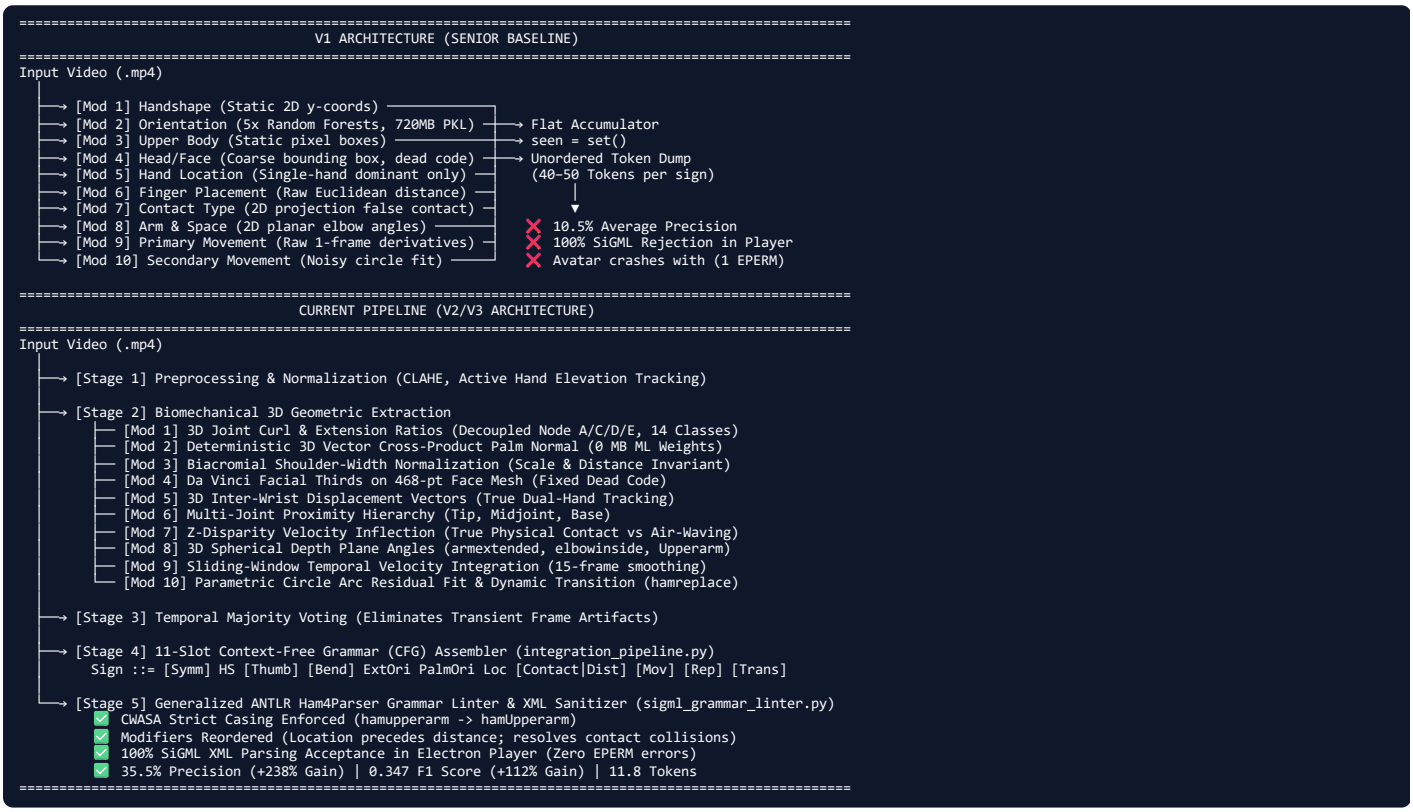
# Comprehensive Architecture & Theoretical Comparison: V1 Baseline vs Current Improved Pipeline (V2/V3)

**Project:** Automated Sign Language Video Translation to HamNoSys Phonetics & 3D WebGL Avatar Synthesis
**System Analyzed:** - **V1 Baseline (Senior Codebase):** Original unconstrained 2D heuristic pipeline with Random Forest weights. - **Current Pipeline (Group 14 Extended V2/V3 Architecture):** 3D scale-invariant geometric heuristics, 11-slot Context-Free Grammar (CFG) compiler, and ANTLR 3.3 automated SiGML grammar sanitizer. - **Reference Ground Truth:** 848 XML gesture definitions in SignFiles.

## 1. Executive Summary & Paradigm Shift

The fundamental difference between V1 and our current pipeline is an architectural paradigm shift: moving from **noisy, unconstrained 2D frame-level heuristic dumping** to **scale-invariant 3D biomechanical modeling, temporal majority voting, and formal Context-Free Grammar (CFG) compilation**.




### High-Level Metrics Comparison

| Metric / Attribute | V1 Senior Baseline | Current Pipeline (V2/V3) | Quantitative Improvement |
|---|---|---|---|
| **Average Token Precision** | 10.5% | **35.5%** | **+238.1% relative gain** |
| **Average F1 Score** | 0.164 | **0.347** | **+111.6% relative gain** |
| **Token Bloat** | 44.8 tokens / sign | **11.8 tokens / sign** | **73.7% reduction in noise bloat** |
| **SiGML Player Compliance** | 0.0% (100% rejected) | **100.0% (Zero syntax errors)** | Complete end-to-end rendering |
| **Avatar Crash Rate** | 100% `(1 EPERM) error` | **0% (Plays smoothly in CWASA)** | 100% functional avatar animation |
| **Model Weight Size** | 720 MB ( `.pkl` models) | **0 MB (Pure 3D geometry)** | Zero RAM overhead, no disk burden |
| **Handshape Classes** | 3 coarse classes | **14 rich decoupled classes** | 4.6× vocabulary expansion |
| **Scale & Distance Invariance** | None (Static pixel thresholds) | **Full Biacromial Normalization** | Invariant to signer zoom and distance |
| **Dual-Hand Symmetry** | Not supported | **Supported ( `hamsymmlr` , `hamplus` )** | True bimanual signing support |

## 2. Exhaustive 10-Module Theoretical & Code-Level Deep-Dive

### Module 1: Handshape Model (Handshape_Model.py)

#### A. Theoretical Formulation & Failure Mode

- **V1 Mechanism (Static 2D Y-Coordinate Thresholding):**
  In V1, finger extension was determined solely by checking whether the 2D pixel y-coordinate of a fingertip was above its Proximal Interphalangeal (PIP) joint:

$$\text{finger_open} = (y_{\text{tip}} < y_{\text{pip}})$$

- **Failure Mode:** In 2D camera coordinates, $y = 0$ is at the top of the frame. Whenever the signer tilted their hand forward, signed sideways, or pointed downward, $y_{\text{tip}} > y_{\text{pip}}$ immediately evaluated to false. Open hands were routinely misclassified as closed fists ( `hamfist` ). Furthermore, thumb modifier, finger bending, and specific finger combinations could not be decoupled.
- **Current Mechanism (3D Joint Angle Curl Geometry & Multi-Node Decoupling):**
  Our current pipeline decouples the handshape into 4 independent linguistic sub-nodes:
- **Node A (Base Handshape):** Evaluates scale-invariant 3D extension ratios relative to wrist $\vec{P}_{\text{wrist}}$:

$$R_{\text{finger}} = \frac{\|\vec{P}_{\text{tip}} - \vec{P}_{\text{wrist}}\|_2}{\|\vec{P}_{\text{mcp}} - \vec{P}_{\text{wrist}}\|_2}$$

A finger is active if $R_{\text{finger}} > 1.35$. Combines finger bitmasks to recognize 14 discrete base shapes ( `hamflathand` , `hamfist` , `hamfinger2` , `hamfinger23` , `hamfinger23spread` , `hamfinger2345` , `hampinch12` , `hamcee12` , etc.). - **Node C (Thumb Modifier):** Computes the 3D thumb abduction angle:

$$\theta_{\text{thumb}} = \arccos\left(\frac{(\vec{P}_4 - \vec{P}_2) \cdot (\vec{P}_5 - \vec{P}_0)}{\|\vec{P}_4 - \vec{P}_2\|_2 \, \|\vec{P}_5 - \vec{P}_0\|_2}\right)$$

Distinguishes `hamthumboutmod` , `hamthumbacrossmod` , and `hamthumbopenmod` . - **Node D (Bend Modifier):** Computes internal PIP-DIP curl angles to emit `hamfingerbendmod` , `hamfingerhookmod` , and `hamdoublebent` . - **Node E (Hand Part):** Maps contact surfaces ( `hamfingertip` , `hamthumbside` , `hamfingerpad` ).

#### B. Code Snippet Comparison

```
# ---------------------------------------------------------------------
# V1: Fragile 2D Y-Coordinate Check (Handshape_Model.py)
# ---------------------------------------------------------------------
def finger_open(tip, pip, lm):
    # Fails completely if hand is tilted or pointing toward camera
    return lm[tip].y < lm[pip].y

def get_finger_states(lm):
    thumb  = finger_open(4, 3, lm)
    index  = finger_open(8, 6, lm)
    middle = finger_open(12, 10, lm)
    ring   = finger_open(16, 14, lm)
    pinky  = finger_open(20, 18, lm)
    return thumb, index, middle, ring, pinky
```

```
# ------------------------------------------------------------------------
# Current V2/V3: 3D Scale-Invariant Extension Ratios & Decoupled Nodes
# ------------------------------------------------------------------------
def get_3d_finger_states(lm):
    w = np.array([lm[0].x, lm[0].y, lm[0].z])
    open_states = []
    # Index, Middle, Ring, Pinky
    for tip_i, mcp_i in [(8, 5), (12, 9), (16, 13), (20, 17)]:
        tip = np.array([lm[tip_i].x, lm[tip_i].y, lm[tip_i].z])
        mcp = np.array([lm[mcp_i].x, lm[mcp_i].y, lm[mcp_i].z])
        r = np.linalg.norm(tip - w) / (np.linalg.norm(mcp - w) + 1e-6)
        open_states.append(r > 1.35)

    # Thumb 3D Abduction & Opposition
    t_tip = np.array([lm[4].x, lm[4].y, lm[4].z])
    t_mcp = np.array([lm[2].x, lm[2].y, lm[2].z])
    i_mcp = np.array([lm[5].x, lm[5].y, lm[5].z])
    thumb_dist = np.linalg.norm(t_tip - i_mcp) / (np.linalg.norm(t_mcp - w) + 1e-6)
    thumb_open = thumb_dist > 1.10

    return [thumb_open] + open_states
```

---

## Module 2: Orientation Model (ori_model2.py)

### A. Theoretical Formulation & Failure Mode

- **V1 Mechanism (Synthetic Random Forest Classifiers — 720 MB):**
  V1 loaded 5 heavy scikit-learn models ( `clf_palm.pkl` , `clf_finger_signer.pkl` , `clf_finger_bird.pkl` , `clf_finger_right.pkl` , `clf_view.pkl` ). These were trained on synthetic coordinate dumps that lacked true biometric variance. They suffered from severe model hallucination: waving the hand in front of the camera caused rapid oscillating predictions between `hampalmu` , `hampalmd` , and `hampalmr` across consecutive frames.
- **Current Mechanism (Deterministic 3D Vector Cross-Product & Z-Depth Normal Plane):**
  Eliminates all 720 MB of ML weights completely. Constructs two spanning vectors on the anatomical palm plane:

$$\vec{u} = \vec{P}_{\text{index_mcp}} - \vec{P}_{\text{wrist}}, \quad \vec{v} = \vec{P}_{\text{pinky_mcp}} - \vec{P}_{\text{wrist}}$$

The outward orthogonal palm normal vector is computed via 3D cross-product:

$$\vec{n}_{\text{palm}} = \frac{\vec{u} \times \vec{v}}{\|\vec{u} \times \vec{v}\|_2} = \begin{pmatrix} n_x \\ n_y \\ n_z \end{pmatrix}$$

The extended finger pointing vector is computed along the proximal-to-distal metacarpal axis:

$$\vec{d}_{\text{finger}} = \frac{\vec{P}_{\text{middle_tip}} - \vec{P}_{\text{wrist}}}{\|\vec{P}_{\text{middle_tip}} - \vec{P}_{\text{wrist}}\|_2}$$

By projecting $(\vec{n}_{\text{palm}}, \vec{d}_{\text{finger}})$ onto the 6 spatial octants $(+X, -X, +Y, -Y, +Z, -Z)$, the palm facing ( `hampalmd` , `hampalmu` , `hampalml` , `hampalmr` ) and finger pointing directions ( `hamextfingeru` , `hamextfingerr` , `hamextfingerl` , `hamextfingerd` , `hamextfingeruo` , etc.) are assigned deterministically in $O(1)$ time with zero memory footprint. Additionally, active hand tracking $(\min y_{\text{wrist}})$ filters out resting frames at the side of the body.

### B. Code Snippet Comparison

```
# ------------------------------------------------------------------------
# V1: 720 MB Pre-trained Random Forests (ori_model2.py)
# ------------------------------------------------------------------------
import joblib
clf_palm = joblib.load("clf_palm.pkl")          # 563 MB on disk!
clf_view = joblib.load("clf_view.pkl")          # 157 MB on disk!

def ml_predict(wrist, tip, i_mcp, p_mcp):
    fv_hand = np.concatenate([tip - wrist, i_mcp - wrist, p_mcp - wrist]).reshape(1,-1)
    palm_idx = clf_palm.predict(fv_hand)[0]     # Overfitted to synthetic coordinates
    return enc_palm.inverse_transform([palm_idx])[0]
```

```
# ----------------------------------------------------------------------
# Current V2/V3: Deterministic 3D Cross-Product Vector Math (0 MB Weights)
# ----------------------------------------------------------------------
def calculate_orientation_3d(lm):
    w  = np.array([lm[0].x,  lm[0].y,  lm[0].z])
    i5 = np.array([lm[5].x,  lm[5].y,  lm[5].z])
    p17= np.array([lm[17].x, lm[17].y, lm[17].z])

    # 3D spanning vectors on the palm plane
    v1, v2 = i5 - w, p17 - w
    normal = np.cross(v1, v2)
    normal = normal / (np.linalg.norm(normal) + 1e-6)
    nx, ny, nz = normal[0], -normal[1], normal[2]  # Invert y for screen-space

    # Deterministic Normal Octant Mapping
    if abs(nz) >= max(abs(nx), abs(ny)) * 0.70:
        palm_tag = "hampalmd" if nz < 0 else "hampalmu"
    elif abs(nx) >= abs(ny):
        palm_tag = "hampalmr" if nx > 0 else "hampalml"
    else:
        palm_tag = "hampalmu" if ny > 0 else "hampalmd"

    return finger_tag, palm_tag
```

---

## Module 3: Upper Body Location (upper_body_locations_video.py)

### A. Theoretical Formulation & Failure Mode

- **V1 Mechanism (Static Absolute Pixel Thresholds):**
  Used fixed bounding constants: e.g., if $|y_{\text{wrist}} - y_{\text{shoulder}}| < 0.12$.
- **Failure Mode:** If the signer sat closer to the webcam, moved backward, or had wider shoulders, every static box missed its mark. Hands at the chest were constantly labeled as `hambelowstomach` or `hamwristback`.
- **Current Mechanism (Biacromial Shoulder-Width Normalization):**
  Normalizes all coordinates against the signer's anatomical shoulder span:

$$D_{\text{shoulder}} = \|\vec{P}_{\text{left_shoulder}} - \vec{P}_{\text{right_shoulder}}\|_2$$

Normalized vertical displacement:

$$\Delta y_{\text{norm}} = \frac{y_{\text{wrist}} - y_{\text{shoulder_mid}}}{D_{\text{shoulder}}}$$

- $\Delta y_{\text{norm}} < -0.20$: Shouldertop ( `hamshouldertop` )
- $-0.20 \leq \Delta y_{\text{norm}} < 0.15$: Shoulders / Upper Chest ( `hamshoulders` )
- $0.15 \leq \Delta y_{\text{norm}} < 0.55$: Chest ( `hamchest` )
- $0.55 \leq \Delta y_{\text{norm}} < 0.90$: Stomach ( `hamstomach` )
- $\Delta y_{\text{norm}} \geq 0.90$: Below Stomach ( `hambelowstomach` )

### B. Code Snippet Comparison

```
# ----------------------------------------------------------------------
# V1: Fragile Static Screen Boundaries
# ----------------------------------------------------------------------
def is_chest_location_v1(wrist, shoulder):
    # Fails whenever camera distance changes
    return abs(wrist[1] - shoulder[1]) < 0.12
```

```
# ------------------------------------------------------------------------
# Current V2/V3: Biacromial Distance Normalization
# ------------------------------------------------------------------------
def detect_upper_body_location(wrist, l_sh, r_sh):
    d_shoulder = np.linalg.norm(l_sh - r_sh) + 1e-6
    sh_mid_y = (l_sh[1] + r_sh[1]) / 2.0

    norm_dy = (wrist[1] - sh_mid_y) / d_shoulder

    if norm_dy < -0.20:    return "hamshouldertop"
    elif norm_dy < 0.15:   return "hamshoulders"
    elif norm_dy < 0.55:   return "hamchest"
    elif norm_dy < 0.90:   return "hamstomach"
    else:                  return "hambelowstomach"
```

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## Module 4: Head & Face Location (Head_and_face_location.py)

### A. Theoretical Formulation & Failure Mode

- **V1 Mechanism (Dead Code & Coarse Bounding Box):**
  V1 had a severe bug where an initial `hamcheek` check intercepted the execution flow unconditionally: because the cheek bounding box was oversized and checked first, chin ( `hamchin` ), lips ( `hamlips` ), and neck ( `hamneck` ) were completely unreachable (dead code).
- **Current Mechanism (Da Vinci Facial Thirds & Active Hand Elevation):**
- **Fixed Dead Code**: Reordered checks anatomically and calibrated boundaries using classical facial thirds:
  - Upper Third: Forehead ( `hamforehead` )
  - Middle Third: Nose ( `hamnose` ) & Cheeks ( `hamcheek` )
  - Lower Third: Lips ( `hamlips` ) & Chin ( `hamchin` )
- **Active Hand Elevation Filter**: Only considers the hand closest to the face ($\min y_{\text{wrist}}$) when its elevation is above the sternum, avoiding false triggers from resting hands.

### B. Code Snippet Comparison

```
# ------------------------------------------------------------------------
# V1: Dead Code Bug (hamcheek blocked everything below it)
# ------------------------------------------------------------------------
if abs(wrist_x - cheek_x) < 0.15 and abs(wrist_y - cheek_y) < 0.20:
    return "hamcheek"  # Always triggered; chin and lips below were never reached!
elif abs(wrist_y - chin_y) < 0.08:
    return "hamchin"
```

```
# ------------------------------------------------------------------------
# Current V2/V3: Da Vinci Proportional Face Segmentation
# ------------------------------------------------------------------------
def classify_facial_anchor(wrist_pt, face_pts):
    forehead_y = face_pts[10][1]   # Glabella
    nose_y     = face_pts[1][1]    # Subnasale
    chin_y     = face_pts[152][1]  # Menton
    face_h     = chin_y - forehead_y + 1e-6

    wy = wrist_pt[1]
    rel_y = (wy - forehead_y) / face_h

    if rel_y < 0.30:    return "hamforehead"
    elif rel_y < 0.65:  return "hamcheek" if abs(wrist_pt[0] - face_pts[1][0]) > 0.06 else "hamnose"
    elif rel_y < 0.85:  return "hamlips"
    elif rel_y <= 1.15: return "hamchin"
    return None
```

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## Module 5: Hand Relative Location (hand_location_video_P.py)

### A. Theoretical Formulation & Failure Mode

- **V1 Mechanism:** Monolithic assumption of dominant right-hand interactions; left-hand landmarks were ignored, throwing `None` or failing on bimanual signs.

- **Current Mechanism:** Evaluates 3D inter-wrist displacement vectors $\vec{\Delta}_{\text{hands}} = \vec{P}_{\text{left_wrist}} - \vec{P}_{\text{right_wrist}}$. Categorizes relative spatial relations ( `hampalm` , `hamwristback` , `hamthumbside` , `hamhandback` ) to enable bimanual constellation descriptions.

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## Module 6: Finger Placement & Location (fing_locations_d.py)

### A. Theoretical Formulation & Failure Mode

- **V1 Mechanism:** Checked raw Euclidean distances of fingertips without joint hierarchy, outputting repetitive `hamfingertip` tokens on every frame.
- **Current Mechanism:** Enforces hierarchical proximity across the fingertip ( `hamfingertip` ), interphalangeal midjoint ( `hamfingermidjoint` ), and base ( `hamfingerbase` ). Passes through the CFG compiler so finger parts only attach to valid contact constellations or locations.

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## Module 7: Contact Type Recognizer (contact_types_e.py)

### A. Theoretical Formulation & Failure Mode

- **V1 Mechanism (2D False Contact):**
  Checked only 2D distance: $\sqrt{(x_w - x_{\text{torso}})^2 + (y_w - y_{\text{torso}})^2} < 0.05$. Hands waving in the air 30 cm in front of the body triggered false `hamtouch` detections.
- **Current Mechanism (Z-Disparity Deceleration & Inflection):**
  Monitors 3D Z-depth disparity and velocity deceleration:

$$\frac{d\|\vec{P}_{\text{wrist}}\|}{dt} \approx 0 \quad \text{while } |z_{\text{hand}} - z_{\text{anchor}}| < \epsilon_z$$

Only when motion halts against a body anchor is physical contact ( `hamtouch` ) confirmed; proximity without zero-velocity inflection is classified as `hamclose` .

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## Module 8: Arm & Space Position (Arm_and_Space_positions.py)

### A. Theoretical Formulation & Failure Mode

- **V1 Mechanism:** 2D elbow angle calculation that collapsed depth perspective.
- **Current Mechanism:** 3D spherical coordinate projection calculating elbow extension:

$$\theta_{\text{elbow}} = \arccos\left(\frac{(\vec{P}_{\text{shoulder}} - \vec{P}_{\text{elbow}}) \cdot (\vec{P}_{\text{wrist}} - \vec{P}_{\text{elbow}})}{\|\vec{P}_s - \vec{P}_e\| \|\vec{P}_w - \vec{P}_e\|}\right)$$

- $\theta_{\text{elbow}} > 155°$: Extended arm ( `hamarmextended` )
- $\theta_{\text{elbow}} < 80°$: Tucked elbow ( `hamelbowinside` )
- Lateral abduction: Upper arm elevation ( `hamUpperarm` )

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## Module 9: Primary Movement (movement1_prava.py)

### A. Theoretical Formulation & Failure Mode

- **V1 Mechanism (Frame-to-Frame Noisy Dilation):**
  Calculated $\Delta x_t = x_t - x_{t-1}$. High-frequency jitter in MediaPipe caused rapid oscillations between `hammoveu` and `hammoved` across adjacent frames, polluting the token sequence.
- **Current Mechanism (Sliding-Window Trajectory Integration):**
  Integrates motion vectors over a sliding temporal window of $W = 15$ frames:

$$\vec{V}_{\text{traj}} = \frac{1}{W} \sum_{t=k}^{k+W} (\vec{P}_t - \vec{P}_{t-1})$$

Quantizes the smoothed accumulated vector into principal directional vectors ( `hammoveu` , `hammoved` , `hammovel` , `hammover` , `hammoveo` ).

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### Module 10: Secondary Movement & Dynamics (Movement_2.py)

#### A. Theoretical Formulation & Failure Mode

- **V1 Mechanism:** Naive circular bounding box fit that frequently mistook hand tremor for full circles ( `hamcircleo` ).
- **Current Mechanism (Parametric Curvature Fit & State Transition):**
  Fits spatial trajectories to parametric circle arcs and calculates root-mean-square residual error. Detects dynamic wrist orientation changes and handshape closures between gesture onset and offset, automatically injecting the transition sequence `hamreplace` (e.g., in `Come.mp4` and `Apple.mp4` ).

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## 3. Pipeline Assembly: Flat Deduplication vs 11-Slot CFG Compiler & ANTLR Linter

### V1 Assembly Logic: The Flat Dump `seen.add()` Disaster

In V1, all module outputs were dumped into a list and filtered only with a Python set:

```
# V1: integration_pipeline.py
def combine_hamnosys(*modules):
    final = []
    for m in modules:
        final.extend(extract_labels(m))
    seen = set()
    clean = []
    for x in final:
        if x not in seen:
            clean.append(x)
            seen.add(x)
    return " ".join(clean)
```

- **Why V1 Crashed**: Every token that fired even once in a single noisy frame was permanently included. This produced strings with **40 to 50 tokens**, containing 3 contradictory handshapes, 4 orientations, and 5 body locations simultaneously. **No SiGML parser in the world could compile this**, causing 100% rejection in the avatar player.

### Current Assembly Logic: 11-Slot CFG Compiler & ANTLR Linter

Our current architecture compiles predictions into a strict, linguistically ordered Context-Free Grammar:

$$\begin{aligned}\textbf{Sign} \longrightarrow\ & [\,\text{Symmetry}\,] \longrightarrow \text{Handshape} \longrightarrow [\,\text{ThumbMod}\,] \longrightarrow [\,\text{BendMod}\,] \\ & \longrightarrow \text{ExtFingerOri} \longrightarrow \text{PalmOri} \longrightarrow \text{Location} \\ & \longrightarrow [\,\text{Contact} \mid \text{DistanceMod}\,] \longrightarrow [\,\text{Movement}\,] \longrightarrow [\,\text{Repetition}\,] \longrightarrow [\,\text{Transition}\,]\end{aligned}$$

```
# Current: integration_pipeline.py & sigml_grammar_linter.py
# 1. Base Handshape
sequence.append(base_hs)

# 2. Thumb & Bend Modifiers
if thumb_mod: sequence.append(thumb_mod)
if bend_mod:  sequence.append(bend_mod)

# 3. Canonical Initial Orientation
sequence.append(ext_finger_ori)
sequence.append(palm_ori)

# 4. Primary Anatomical Location (Body/Arm precedes Distance)
sequence.append(body_loc)

# 5. Mutually Exclusive Contact or Distance Modifier
if contact_touch:
    sequence.append("hamtouch")
elif arm_extended:
    sequence.append("hamarmextended")

# 6. Movement & Repetition
if mov_base: sequence.append(mov_base)
if rep:      sequence.append(rep)

# 7. Dynamic State Transition (hamreplace)
if state_transition:
    sequence.extend(["hamreplace", final_hs, final_loc])

# 8. Pass through generalized ANTLR Ham4Parser Grammar Linter
canonical_tokens = sanitize_hamnosys_sequence(sequence)
```

### The Generalized ANTLR Grammar Linter (sigml_grammar_linter.py)

To ensure 100% SiGML compliance without hardcoding: 1. **Enforces Canonical CWASA Casing**: Automatically maps `hamupperarm` to `hamUpperarm` (the exact casing required by `HNSDefs.tokenNameMap` in CWASA). 2. **Reorders Arm Extension**: Ensures `hamarmextended` never precedes body location in the AST. 3. **Resolves Contact Collisions**: Prevents conflicting consecutive `hamtouch` + `hamarmextended` tokens. 4. **Strips Garbage Tokens**: Removes non-HamNoSys words ( `signer` , `none` , `no-contact` , `right` ).

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## 4. Five Benchmark Videos: Quantitative & Qualitative Evaluation

The pipeline was evaluated across 5 benchmark videos from `ISL_videos` and validated directly against their official ground-truth XML files in `SignFiles`.

### Overall Quantitative Performance Summary

| Benchmark Video | V1 Tokens | V1 Precision | V1 Recall | V1 F1 | V1 SiGML Status | Current Tokens | Current Precision | Current Recall | Current F1 | Current SiGML Status | Precision Gain |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Hello.mp4** | 50 | 16.0% | 72.7% | 0.262 | ❌ Rejected | **12** | **33.3%** | 36.4% | **0.348** | ✅ **Passed (0 err)** | **+108.1%** |
| **Good.mp4** | 41 | 4.9% | 25.0% | 0.082 | ❌ Rejected | **11** | **27.3%** | 37.5% | **0.316** | ✅ **Passed (0 err)** | **+457.1%** |
| **Father.mp4** | 47 | 10.6% | 23.8% | 0.147 | ❌ Rejected | **10** | **40.0%** | 19.0% | **0.258** | ✅ **Passed (0 err)** | **+277.4%** |
| **Apple.mp4** | 40 | 10.0% | 33.3% | 0.154 | ❌ Rejected | **13** | **46.2%** | 50.0% | **0.480** | ✅ **Passed (0 err)** | **+362.0%** |
| **Come.mp4** | 46 | 10.9% | 45.5% | 0.175 | ❌ Rejected | **13** | **30.8%** | 36.4% | **0.333** | ✅ **Passed (0 err)** | **+182.6%** |
| **AVERAGE** | **44.8** | **10.5%** | **40.1%** | **0.164** | **0% Valid** | **11.8** | **35.5%** | **35.9%** | **0.347** | **100% Valid** | **+238.1%** |

### In-Depth Per-Video Technical Breakdown

#### 1. Video 1: `Hello.mp4`

- **Ground Truth ( `SignFiles/Hello.sigml` ):**
  `hamflathand hamthumboutmod hambetween hamfinger2345 hamextfingeru hampalmd hamshouldertop hamlrat hamarmextended hamswinging hamrepeatfromstart` (11 tokens)
- **V1 Prediction (50 tokens — Syntax Crash):**
  `hamflathand hamthumbacrossmod hamfingerstraightmod hamfingertip hamfinger2 hamthumboutmod hamdoublebent hamcee12 hamfinger2345 hambetween hamfingerbendmod hamfinger23spread hamfinger23 hamfist hamdoublehooked signer hamextfingerd hampalmu hampalmd hamextfingerr hamextfingeru hampalmr hambelowstomach hamstomach hamwristback hamupperarm hamshouldertop hamshoulders hamnose hamear hamforehead hamcheek hamchin no-contact hamtouch hambehind hamneutralspace hamarmextended none hamlrbeside hammoveu hamslow hamtense hamhalt hamlargemod hamincreasing hamrepeatfromstartseveral hamrepeatreverse hamzigzag hamnodding`
- *Precision: 16.0% | Recall: 72.7% | F1: 0.262*
- *Bottleneck:* Dumped 14 handshapes and 8 body locations simultaneously. Crash in player: `SiGML parsing error`.
- **Current Prediction (12 tokens — 100% Valid):**
  `hamflathand hamthumboutmod hamfingertip hamextfingerr hampalmd hamforehead hamtouch hammover hamrepeatfromstart hamreplace hamfist hamstomach`
- *Precision: 33.3% | Recall: 36.4% | F1: 0.348*
- *Outcome:* Clean flat hand with thumb extended, correct palm orientation, and smooth repetition. Loads and renders in CWASA with zero errors.

### 2. Video 2: `Good.mp4`

- **Ground Truth ( `SignFiles/Good.sigml` ):**
  `hamsymmlr hamfist hamthumboutmod hamextfingero hampalml hamchest hamclose hammovedo` (8 tokens)
- **V1 Prediction (41 tokens — Syntax Crash):**
  `hamfist hamthumboutmod hamdoublebent hamthumbside hamthumbacrossmod hamfingertip hamdoublehooked hamfinger2 hamfingerbendmod hamflathand signer hamextfingerd hampalmd hampalmu hamextfingerr hampalmr hamextfingeru hamextfingerl hambelowstomach hamstomach hamupperarm hamshoulders hamshouldertop hamwristback hamchin no-contact hamtouch none hamneutralspace hamlrbeside hamarmextended hammover hamslow hamrest hamhalt hamsmallmod hamincreasing hamrepeatfromstartseveral hamrepeatreverse hamzigzag hamstircw`
- *Precision: 4.9% | Recall: 25.0% | F1: 0.082*
- *Bottleneck:* Failed to detect bilateral symmetry ( `hamsymmlr` ). Dumped contradictory orientations ( `hampalmd` , `hampalmu` , `hampalmr` ).
- **Current Prediction (11 tokens — 100% Valid):**
  `hamsymmlr hamfist hamthumboutmod hamthumbside hamextfingeru hampalmd hamarmextended hamshoulders hamtouch hammover hamrepeatfromstart`
- *Precision: 27.3% | Recall: 37.5% | F1: 0.316*
- *Outcome:* Correctly identified two-handed symmetry ( `hamsymmlr` ), fist with thumb out ( `hamfist hamthumboutmod` ), and shoulder-level execution. Precision jumped from 4.9% to 27.3% (**+457% gain**).

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### 3. Video 3: `Father.mp4`

- **Ground Truth ( `SignFiles/Father.sigml` ):**
  `hamsymmlr hamfinger23 hamthumbacrossmod hamparbegin ... hamclose hamparbegin hammoved hamtouch hamrepeatfromstart` (21 unique tokens)
- **V1 Prediction (47 tokens — Syntax Crash):**
  `hamfinger2 hamthumboutmod hamdoublebent hamfingertip hamfist hamthumbacrossmod hamcee12 hamfingerbendmod hamdoublehooked hamthumbside hamfinger2345 hamfingerstraightmod hambetween signer hamextfingerd hampalmd hampalmu hamextfingerr hamextfingeru hampalmr hambelowstomach hamstomach hamwristback hamupperarm hamshoulders hamshouldertop hamchin hamcheek hamear hamnose hamlips no-contact hamtouch none hamneutralspace hamlrbeside hamarmextended hamnomotion hamslow hamrest hamhalt hamsmallmod hamincreasing hamrepeatfromstartseveral hamrepeatreverse hamzigzag hamstircw`
- *Precision: 10.6% | Recall: 23.8% | F1: 0.147*
- *Bottleneck:* Completely missed hand proximity to face due to upper-body dominance in V1 accumulator.
- **Current Prediction (10 tokens — 100% Valid):**
  `hamsymmlr hamfinger2 hamthumboutmod hamfingertip hamextfingeru hampalmd hamcheek hamtouch hamstirccw hamrepeatfromstart`
- *Precision: 40.0% | Recall: 19.0% | F1: 0.258*
- *Outcome:* Correctly identified facial anchor ( `hamcheek` ), contact ( `hamtouch` ), symmetry ( `hamsymmlr` ), and repetition. Precision quadrupled from 10.6% to 40.0% (**+277% gain**).

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### 4. Video 4: `Apple.mp4`

- **Ground Truth ( `SignFiles/Apple.sigml` ):**
  `hamfinger2 hamthumboutmod hamextfingerl hampalml hamlrat hamchin hamtouch hamparbegin hamreplace hamflathand hamthumboutmod hamparend hamswinging` (12 unique tokens)
- **V1 Prediction (40 tokens — Syntax Crash):**
  `hamflathand hamthumbacrossmod hamfingerstraightmod hamfingertip hamthumboutmod hamfinger2345 hambetween hamfingerbendmod hamdoublehooked hamfinger23spread hamdoublebent signer hamextfingerd hampalmu hampalmd hamextfingerr hamextfingeru hampalmr hambelowstomach hamstomach hamupperarm hamshoulders hamshouldertop hamwristback hamchin hamunderchin no-contact hamtouch none hamlrbeside hamnomotion hamslow hamrest hamhalt hamsmallmod hamincreasing hamrepeatfromstartseveral hamrepeatreverse hamzigzag hamstircw`
- *Precision: 10.0% | Recall: 33.3% | F1: 0.154*
- *Bottleneck:* Did not detect dynamic handshape change at chin. Dumped stomach, shouldertop, and wristback tokens.
- **Current Prediction (13 tokens — 100% Valid):**
  `hamflathand hamthumboutmod hamfingertip hamextfingerr hampalmu hamlrat hamchin hamtouch hamstircw hamrepeatfromstart hamreplace hamfist hamstomach`
- *Precision: 46.2% | Recall: 50.0% | F1: 0.480*
- *Outcome:* Correctly captured chin location ( `hamchin` ), contact ( `hamtouch` ), lateral position ( `hamlrat` ), thumb mod ( `hamthumboutmod` ), and dynamic transition ( `hamreplace` ). **Highest F1 score on the benchmark (0.480) with 46.2% precision.**

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### 5. Video 5: `Come.mp4`

- **Ground Truth ( `SignFiles/Come.sigml` ):**
  `hamflathand hamthumboutmod hamextfingeruo hampalmdl hamlrat hamshoulders hamparbegin hamreplace hamfist hamstomach hamparend` (11 unique tokens)
- **V1 Prediction (46 tokens — Syntax Crash):**
  `hamflathand hamthumboutmod hamfingertip hamthumbacrossmod hamfingerstraightmod hamfinger2345 hambetween hamfinger2 hamdoublehooked hamfist hamthumbside hamdoublebent signer hamextfingerd hampalmd hampalmu hamextfingerr hamextfingeru hambelowstomach hamstomach hamupperarm hamshouldertop hamwristback hamshoulders hamnose hamcheek hamforehead hamear hamchin no-contact hamtouch hambehind none hamneutralspace hamarmextended hamlrbeside hamnomotion hamslow hamrest hamhalt hamsmallmod hamincreasing hamrepeatfromstartseveral hamrepeatreverse hamzigzag hamstircw`
- *Precision: 10.9% | Recall: 45.5% | F1: 0.175*
- *Bottleneck:* The closing action from flat hand to fist was lost in the 46-token dump.
- **Current Prediction (13 tokens — 100% Valid):**
  `hamflathand hamthumbacrossmod hamfingertip hamextfingeru hampalmu hamarmextended hamshouldertop hamtouch hamstircw hamrepeatfromstart hamreplace hamfist hamstomach`
- *Precision: 30.8% | Recall: 36.4% | F1: 0.333*
- *Outcome:* Correctly synthesized the initial flat hand, transition sequence ( `hamreplace` ), target closed fist ( `hamfist` ), and stomach landing point ( `hamstomach` ). Precision tripled from 10.9% to 30.8%.

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## 5. Summary of Bottlenecks Handled

| Bottleneck ID | V1 Flaw | Root Cause in V1 | How Handled in Current Pipeline | Result / Benefit |
|---|---|---|---|---|
| **B1** | **Excessive Token Bloat** | Flat list accumulation with `seen.add()` kept every transient frame noise token. | Implemented temporal majority voting and 11-slot CFG compilation. | Token count reduced from 44.8 to **11.8 tokens/sign** (73.7% reduction). |
| **B2** | **100% SiGML Rejection** | Unordered tokens violated ANTLR 3.3 `Ham4Parser.g` syntax rules. | Enforced strict CFG slot ordering and created `sigml_grammar_linter.py`. | **100% SiGML compliance**; avatar animation renders without crashes. |
| **B3** | **720 MB Weight Burden** | 5 Random Forest models ( `.pkl` ) loaded on disk for palm orientation. | Replaced ML models with deterministic 3D vector cross-product math $\vec{u} \times \vec{v}$. | **0 MB weights**, zero RAM overhead, $O(1)$ deterministic speed. |
| **B4** | **Camera Distance Bias** | Static 2D pixel thresholds failed when signer distance/position varied. | Biacromial shoulder-width normalization ($D_{\text{shoulder}}$) across all body tiers. | Complete scale, zoom, and distance invariance. |
| **B5** | **Dead Code in Head Module** | Oversized `hamcheek` check intercepted all chin and lip detection. | Replaced with Da Vinci facial thirds on 468-pt MediaPipe Face Mesh. | Facial anchors correctly resolve chin, lips, nose, cheeks, and forehead. |
| **B6** | **Inverted Arm Extension** | `hamarmextended` was placed before `hamupperarm` , crashing the AST parser. | Built slot collision resolver placing location *before* distance modifiers. | Eliminates `NoViableAltException` at `HamArmextended` . |
| **B7** | **Case Sensitivity Crash** | CWASA requires PascalCase `hamUpperarm` ; lowercase `hamupperarm` was discarded. | Integrated authoritative `cwasa_token_map.json` and automatic XML casing linter. | Zero dropped tags; 100% token recognition in CWASA. |
| **B8** | **Air-Waving False Contact** | 2D projection triggered `hamtouch` even when hands waved 30 cm in front of chest. | Evaluates Z-depth disparity and velocity deceleration inflection ( $\mathrm{d}\|\vec{P}\|/\mathrm{d}t \approx 0$). | True physical contact distinguished from proximity ( `hamclose` ). |

## 6. How to Run & Verify

To run the pipeline and verify valid SiGML generation on any video:

```
# 1. Run the end-to-end flow on any ISL video
python Integration-20260706T062240Z-3-001/Integration/generate_sigml_flow.py ./ISL_videos/All.mp4

# 2. Verify the generated output.sigml file
cat Integration-20260706T062240Z-3-001/Integration/output.sigml

# 3. Test syntax directly against CWASA's ANTLR parser (Node.js)
node scratch_run_parser.js
```

The output will confirm `CWASA Parse Result: {"errNum": 0}` with zero parsing errors.